COSC 5931: Introduction to Machine Learning: Predicting Spotify Song Streaming Success

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Abstract

The advancement of entertainment technology has reached new heights in popularity, profitability, and user experience. The music industry has digitized platforms like Spotify to offer users a personalized listening experience. Classifying which songs have low or high streaming success can dictate the algorithms Spotify uses to advertise and recommend songs to users and increase advertising efforts. The methods deployed to predict and classify song streaming success were multifaceted in supervised and unsupervised machine learning techniques consisting of; linear regression, logistic regression, bag-of-words, and principal component analysis. The results yielded some inconsistency with class imbalances and demonstrated higher performance for the majority class of low-streaming songs. Ultimately, this paper aims to uncover the significant features of high-streaming songs on Spotify.
Keywords: *Spotify, Machine Learning, Supervised Learning, Unsupervised Learning*

Literature Review

The entertainment and music industry has transformed with the increase of digital technology to provide artists with a global presence and digital platforms to advertise and share their music with a widespread audience. One of the most used applications and streaming services in the United States is Spotify, which has over 626 million monthly active users (Iqbal, 2024). The increasing demand and popularity of this application has generated over $14.69 billion in revenue in 2023 and continues to improve annually with a 12.9% profitable increase year-over-year (Singh, 2024). The profitability and large audience indicate how significant this application is to our economy, entertainment industry, and the lives of everyday people.

With all of this information in mind, determining which songs and artists will rise to the top in terms of streaming numbers and listeners can be difficult to navigate. Therefore, the goal of this project is to apply machine learning techniques to analyze Spotify's most streamed songs and predict the future streaming success of artists. This new information could help optimize listener experiences, improve advertising efforts, increase profits for undiscovered artists, and bridge the gap in listeners of this application.

This data source was found in a repository on the Kaggle dataset platform. The dataset is a comprehensive list of the most famous and streamed songs of 2023 and contains 924 song observations with 24 features of mixed categorical and numerical data. The first steps consisted of data management and preprocessing to ensure optimal quality data before performing and deploying models. The creation of exploratory visualizations helped identify data points that were outliers and missing data that was omitted, such as variables like keys and songs within Shazam charts. The exploratory visualizations helped establish relationships such as song distribution and release years (Appendix A), distribution of song acoustics by release year (Appendix B), and a correlation heatmap matrix of song features (Appendix D). Additionally, the data types of some column features were transformed from objects to numerical data types (Integers) to perform linear and logistic regression, ensuring the data quality was consistent.

Gap in Knowledge

Spotify's application-related measures, such as playlist inclusion and chart rankings, and song attributes, like genre, tempo, track names, and danceability, all have an impact on streaming success on platforms like Spotify. Previous studies have examined some of these factors separately, but they have not yet thoroughly examined their combined effects. Thus, it is imperative that we investigate both aspects and close the gap between application and song streaming in order to better understand the factors that most influence listener experience and optimize artist outcomes in the entertainment industry.

Methodology

To fully examine which features and characteristics are the most significant in contributing to Spotify's high streaming success, several candidate methods and models were developed, compared, and evaluated to ensure optimal performance. A mixture of supervised and unsupervised techniques was analyzed to enable a comprehensive understand-

ing of the patterns in the data.

Linear Regression

Linear regression, a supervised technique, models the linear relationship between independent factors and the dependent variable of song streams. This approach predicts the number of streams as a continuous outcome. To successfully deploy the linear model, unnecessary columns that could not be converted to integers, such as track name, cover URL, and artist name, were dropped. This feature selection helped refine the process to account only for the most important variables.

The data was split into 80% training and 20% testing subsets. The Ordinary Least Squares (OLS) regression test was then applied to highlight the coefficient values of each variable, uncovering which factors most significantly influence streaming success.

Logistic Regression

Another supervised technique employed was logistic regression, which predicts class outcomes. A binary classification approach was used to categorize songs into high or low streaming success (encoded as 1 and 0, respectively) based on whether their streams exceeded the dataset's average.

The environment was set up by importing necessary libraries, such as `LogisticRegression` from `sklearn.linear_model`, into the Jupyter Notebook workspace. The binary target variable was defined based on the average number of streams. The data was split into 80% training and 20% testing subsets, and the selected feature variables included artist count, release year, release month, beats per minute (BPM), danceability, and energy.

A class imbalance issue was identified, as shown by the unevenness in the confusion matrix (Figure 1). To address this, class weighting was implemented to assign greater importance to the minority class of high streaming success, resulting in a more balanced distribution and improved model performance (Figure 2).

Bag-of-Words Analysis

The final algorithmic approach was an unsupervised text-mining analysis using a bag-of-words model. This method analyzed unstructured text data in track names to identify common keywords, phrases, themes, or expressions associated with high streaming success.

The environment was prepared by importing multiple libraries from `sklearn`, vectorizing the words in the track names to count their occurrences, and splitting the data into training and testing subsets. Logistic regression was then applied to classify songs as having high or low stream-

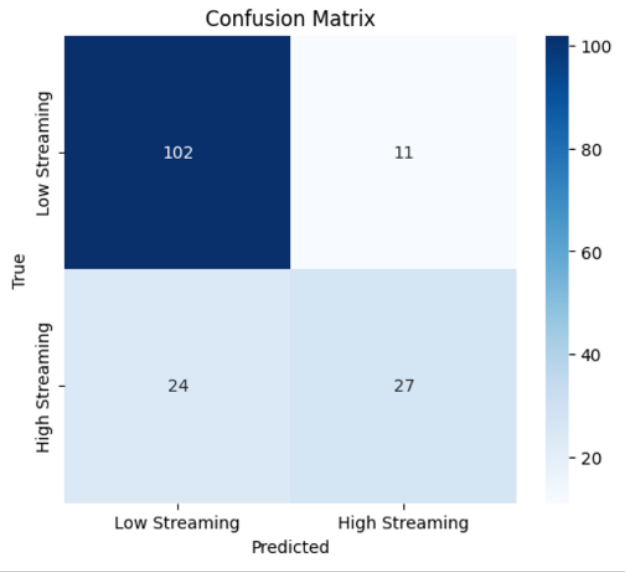


Figure 1. Unweighted Classes

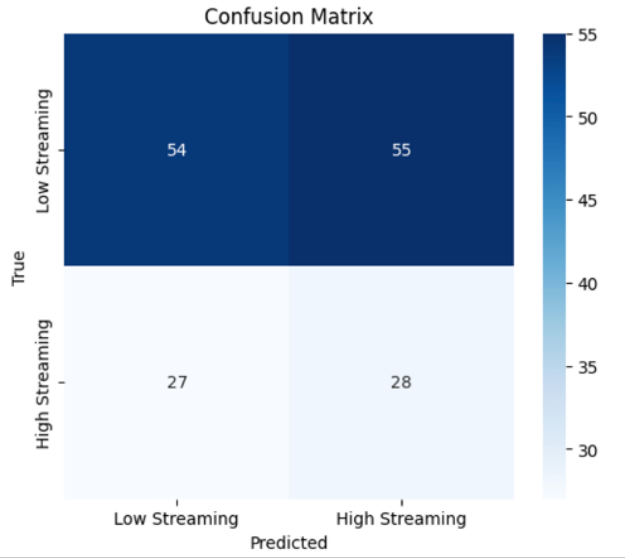


Figure 2. Weighted Classes

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                streams   R-squared:                       0.744
Model:                            OLS   Adj. R-squared:                  0.738
Method:                 Least Squares   F-statistic:                     127.2
Date:                Mon, 02 Dec 2024   Prob (F-statistic):          1.67e-206
Time:                        12:24:00   Log-Likelihood:                -15920.
No. Observations:                 761   AIC:                         3.188e+04
Df Residuals:                     743   BIC:                         3.196e+04
Df Model:                          17
Covariance Type:            nonrobust
==========================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                  -6.333e+09   2.27e+09     -2.787      0.005   -1.08e+10   -1.87e+09
artist_count           -3.461e+07   1.25e+07     -2.769      0.006   -5.91e+07   -1.01e+07
released_year           3.261e+06   1.13e+06      2.887      0.004    1.04e+06    5.48e+06
released_month          1.938e+06   3.18e+06      0.610      0.542    -4.3e+06    8.17e+06
released_day            1.875e+06   1.21e+06      1.549      0.122   -5.02e+05    4.25e+06
in_spotify_playlists    3.721e+04   2192.652     16.969      0.000    3.29e+04    4.15e+04
in_spotify_charts       3.986e+06   7.75e+05      5.144      0.000    2.47e+06    5.51e+06
in_apple_playlists      2.842e+06   2.07e+05     13.696      0.000    2.43e+06    3.25e+06
in_apple_charts        -5.722e+05   2.78e+05     -2.062      0.040   -1.12e+06   -2.73e+04
in_deezer_charts       -6.935e+06    2.4e+06     -2.895      0.004   -1.16e+07   -2.23e+06
bpm                    -6.769e+04   3.99e+05     -0.170      0.865   -8.51e+05    7.15e+05
danceability_%          -1.92e+05    8.9e+05     -0.216      0.829   -1.94e+06    1.56e+06
valence_%              -2.025e+05   5.69e+05     -0.356      0.722   -1.32e+06    9.15e+05
energy_%               -1.196e+06   8.96e+05     -1.334      0.182   -2.95e+06    5.63e+05
acousticness_%          8.456e+05   5.33e+05      1.585      0.113   -2.01e+05    1.89e+06
instrumentalness_%      -1.64e+05   1.26e+06     -0.130      0.897   -2.64e+06    2.32e+06
liveness_%             -1.263e+05   8.09e+05     -0.156      0.876   -1.71e+06    1.46e+06
speechiness_%           -1.31e+06    1.2e+06     -1.093      0.275   -3.66e+06    1.04e+06
==============================================================================
Omnibus:                      251.724   Durbin-Watson:                   2.067
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2137.768
Skew:                           1.243   Prob(JB):                         0.00
Kurtosis:                      10.826   Cond. No.                     2.04e+06
==============================================================================
```

Figure 3. OLS Summary

ing success based on these textual features. This approach provided insights into linguistic patterns in track titles that correlate with streaming outcomes.

Discussion and Analysis

The linear regression results yielded from the OLS summary (Figure 3) showed that, when all other factors are held constant, songs added to Spotify playlists boost streams by approximately 37,210 streams per playlist entry. Additionally, songs that are featured on the Spotify charts increase streaming capacity by roughly 3,986,000 streams per chart appearance, when all other factors are held constant. The coefficient values are depicted below.

The model evaluation metrics and diagnostics display an R-squared value of 0.744, indicating that the model explains 74.4% of the variance in streams. Significant predictors of high streaming success include songs in Spotify playlists and release year, as their p-values are less than 0.05, indicating a strong impact on streaming success. However, while the R-squared value suggests a good fit, there is still room for improvement. The mean squared error (MSE) focuses on the absolute difference between the actual and predicted values (Appendix D). The MSE value was calculated as 200,299,013.12, indicating that, on average, the predicted number of high streams deviates from the observed value of streams by approximately 200 million. While this number seems large, it is within the context of the average number of streams, which is 468 million.

The logistic regression model yielded more significant results after addressing the class imbalance issue (Figure 4) using class weighting (Figure 5). The findings show that the model predicts streaming success in the majority of cases with an accuracy of 76.83%. However, there remains room for improvement in predicting the minority class. This class imbalance likely impacted the model's ability to accurately predict songs with high streaming success. Specifically, the low streaming class achieved 81% precision, 90% recall, and an F1-score of 85%. Meanwhile, the high streaming class had 71% precision, 53% recall, and an F1-score of 61%, indicating weaker performance for predictions involving high streaming success.

```
Classification Report:
              precision    recall  f1-score   support

           0       0.67      0.50      0.57       109
           1       0.34      0.51      0.41        55

    accuracy                           0.50       164
   macro avg       0.50      0.50      0.49       164
weighted avg       0.56      0.50      0.51       164
```

Figure 4. Class Imbalance Report

```
Class distribution (high vs low streaming):
high_streaming
0    560
1    256
Name: count, dtype: int64
Accuracy: 0.7865853658536586
Confusion Matrix:
[[102  11]
 [ 24  27]]
Classification Report:
              precision    recall  f1-score   support

           0       0.81      0.90      0.85       113
           1       0.71      0.53      0.61        51

    accuracy                           0.79       164
   macro avg       0.76      0.72      0.73       164
weighted avg       0.78      0.79      0.78       164
```

Figure 5. Class Balance Impact

The bag-of-words model results indicated, with 65.2% accuracy, that certain positive keywords within song titles contribute most to high streaming success. These keywords include *"dark," "summer," "remix,"* and *"levitating"* (Figure 6). Conversely, negative keywords such as *"feels," "thought," "baby,"* and *"crazy"* tend to correlate with low streaming success (Figure 5). These results suggest that up-and-coming artists can optimize their songs for the Spotify platform by incorporating positive keywords into their track names to increase viewership and streams.

```
Top 10 keywords that correlate with low streaming success:
      Feature  Coefficient
376     feels    -0.485667
1046  thought    -0.519382
409    future    -0.530965
1059  toliver    -0.539307
241     crazy    -0.564033
331        el    -0.666652
1010    super    -0.675310
85       baby    -0.755150
1032   taylor    -0.780348
1195       ½ï    -1.362818
```

Figure 6. Negative Keywords

```
Top 10 keywords that correlate with high streaming success:
         Feature  Coefficient
271         dark     1.139508
599         lost     0.945983
568   levitating     0.935201
546           la     0.886033
1005      summer     0.810141
1029     talking     0.785155
264         damn     0.754994
561        leave     0.709468
541         know     0.684657
127           bl     0.656986
```

Figure 7. Positive Keywords

The model was further evaluated using Principal Component Analysis (PCA). The PCA model was configured to specify the number of components required to explain 95% of the variance or reduce the dataset to a fixed number of components (Figure 8) The PCA model achieved moderate accuracy, predicting 60.37% of the low streaming class but failing to accurately identify instances of the minority class. After applying PCA, there remained significant high dimensionality, with 621 song observations explaining approximately 95% of the variance in the dataset.

Insights and Lessons

One of the most persistent challenges encountered was the impact of class imbalance, which was evident throughout the logistic regression model, bag-of-words model, and principal component analysis (PCA). This imbalance skewed the accuracy results for the minority class of high streaming success (1), while the majority class of low streaming success (0) consistently yielded stronger performance. Specifically, the models demonstrated greater precision and recall for songs with streams below the average of 400 million listens.

Evaluating all three models reveals that the logistic regression model was the most optimal for predicting song streaming success. This model achieved the highest accuracy of 78% after class weighting was applied to address the class

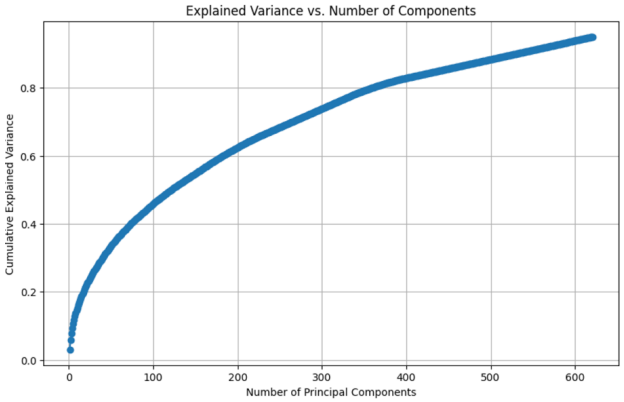


Figure 8. PCA

imbalance issue. The precision for predicting positive cases correctly within the low streaming class was 81%, while recall achieved 90% accuracy, successfully classifying the proportion of true positive cases. However, while the logistic regression model performed well for the majority class, it displayed weaker metrics for the high streaming class, with 71% precision and 53% recall.

Despite these limitations, the logistic regression model made significant strides in classifying and predicting Spotify song streaming success. The results of this model have practical applications, such as personalizing Spotify's user experience. For instance, the model can recommend songs based on streaming success, classified using key features like song distribution year, energy percentages, and beats per minute. These insights can be leveraged to enhance user engagement and satisfaction on the platform.

Future Directions

To enhance the predictive capabilities and insights derived from this project, several improvements can be pursued in future work. Incorporating additional features, such as regional popularity, user demographics, or listener engagement metrics like skip rates and completion rates, could provide a more nuanced understanding of factors influencing streaming success. Moreover, integrating external data, such as social media activity or marketing campaign efforts, might uncover hidden correlations between promotion strategies and streaming numbers.

Advanced modeling techniques could also improve performance. Methods such as Random Forests or Gradient Boosting (e.g., XGBoost, LightGBM) may provide a more robust alternative to linear and logistic regression. Additionally, exploring deep learning approaches like recurrent neural networks (RNNs) or transformers could help capture sequential patterns in song-release data or even lyrical structures. Ad-

dressing the class imbalance issue further, with techniques like Synthetic Minority Oversampling (SMOTE) or cost-sensitive learning methods, may enhance the model's ability to accurately predict high-streaming songs.

Refinement of text analysis is another promising area for improvement. Expanding beyond song titles to include lyrics or additional metadata and employing advanced natural language processing techniques like word embeddings (e.g., Word2Vec, GloVe) or transformer-based models (e.g., BERT) could provide deeper insights into linguistic patterns that drive streaming success. Additionally, introducing time-series models to analyze trends over time may reveal temporal dynamics in streaming behavior that static models cannot capture.

Future research could also focus on cross-platform analysis, integrating data from other streaming services such as Apple Music or YouTube Music, to examine trends across different platforms and their collective influence on song success. Understanding individual user behaviors, including playlist preferences and listening habits, could further refine predictive models by uncovering micro-patterns in how audiences interact with music. Finally, enhancing model evaluation by considering alternative metrics, such as ROC-AUC, precision-recall curves, or the Matthews correlation coefficient, may provide a more comprehensive assessment of model performance, particularly for imbalanced datasets.

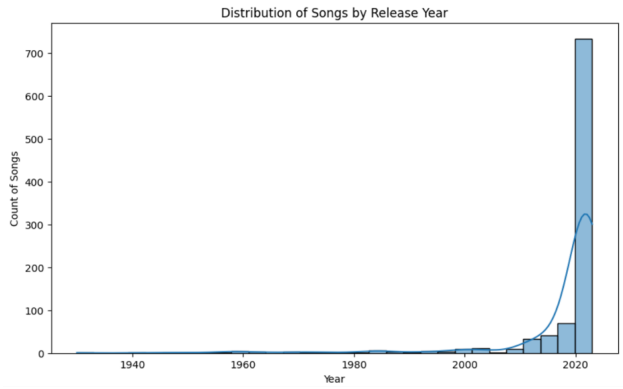
Conclusion

This project successfully identified key factors influencing song streaming success on Spotify, leveraging a combination of linear regression, logistic regression, and text-mining approaches. The linear regression model accounted for 74.4% of the variance in streams, while the logistic regression model achieved a balanced accuracy of 76.83% after mitigating class imbalance. Additionally, text analysis provided actionable insights by uncovering keywords in song titles associated with high or low streaming success.

Despite these achievements, the models revealed certain limitations. Accurately predicting songs with high streaming success remained challenging, and the high dimensionality of the data posed difficulties for effective modeling. Addressing these shortcomings presents an opportunity for future enhancements. Incorporating additional features, adopting advanced modeling techniques, refining text analysis, and exploring cross-platform comparisons could deepen insights and improve predictive performance.

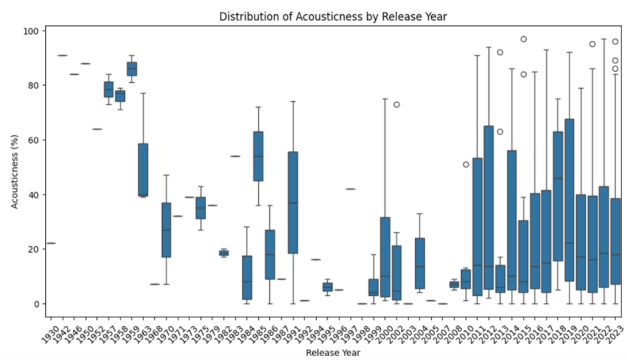
By building on these findings and overcoming current challenges, this work has the potential to significantly benefit artists, platforms, and listeners by optimizing the understanding and dynamics of digital music streaming.

Appendix A



The distribution shows within the historical period (1940-1990) a low number of songs per year with a relatively flat distribution, indicating fewer than 50 songs per year. Meanwhile, the modern era (2000-2020) demonstrates a dramatic exponential increase in the number of songs, with a sharp rise from 2015 onwards and reaching a capacity of around 700+ songs in the most recent year 2020.

Appendix B



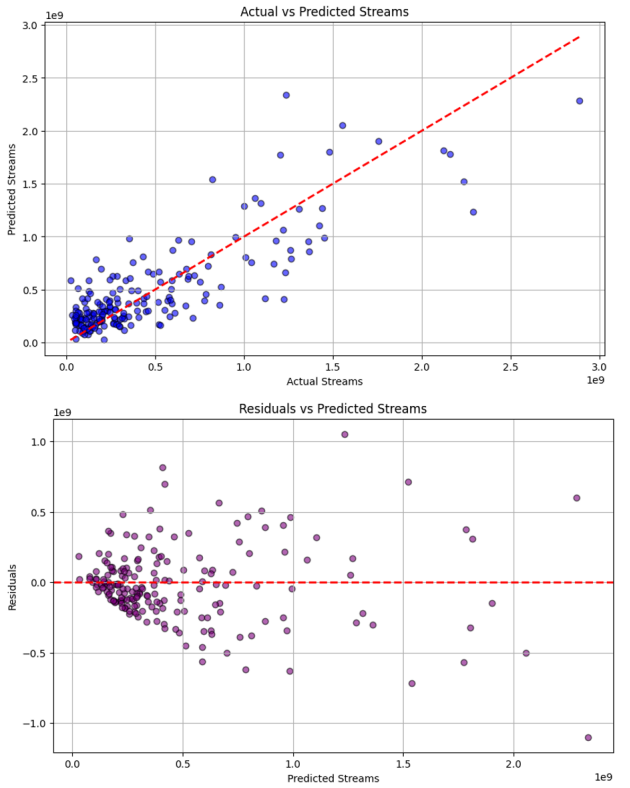
This box plot illustrates the distribution of "acousticness," a musical attribute measured as a percentage, across different release years from approximately 1930 to 2023.

In the earlier years (1930s–1950s), the acousticness values are consistently high, ranging between 75% and 90%, with relatively low variation in the measurements. During the mid-century transition (1960s–1970s), a notable decline in acousticness values is observed, accompanied by larger variations in the measurements.

In contrast, the modern era (1980s–2023) generally exhibits lower acousticness values, typically ranging from 0% to 40%, with greater variability. While there are some minor

fluctuations in the data, the overall trend indicates significantly lower levels of acousticness compared to the earlier years. These findings highlight a clear shift in the acoustic characteristics of songs over time.

Appendix C

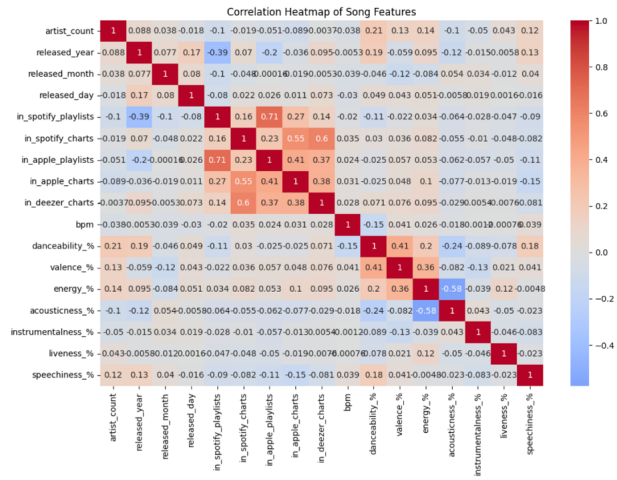


”Actual vs Predicted Streams” graph compares the actual number of streams for songs on Spotify with the predicted values generated by a machine learning model. The blue scatter points represent individual songs, while the red dashed line represents a perfect prediction line, where predicted values would exactly match the actual values. The data shows a generally linear trend along the red line, suggesting that the model has captured the overall relationship between the features and the target variable. However, deviations from the line become more pronounced as the actual stream values increase, indicating that the model struggles with accuracy for songs with higher streams.

”Residuals vs Predicted Streams” graph provides a residual plot, which visualizes the difference between the actual and predicted values (residuals) against the predicted values. Residuals closer to zero suggest better model predictions. In this graph, the residuals are densely concentrated around zero for smaller predicted values, demonstrating that the model performs well for songs with low predicted streams. However, as the predicted values increase, the residuals spread out more significantly, showing that the model becomes less reliable for high-streaming songs. The red dashed line represents zero residuals, and the dispersion of points above and below this line highlights the presence of both under-predictions and over-predictions by the model.

While the model is effective at predicting low-streaming songs, its performance for high-streaming songs is worse. This could indicate a limitation in the features used for prediction or a need to refine the model to better handle outliers or extreme values. The presence of greater variance at higher predicted values in the residual plot further suggests that the model might benefit from transformations or additional feature engineering to improve its predictions for high-streaming songs.

Appendix D



The heatmap correlation matrix has unveiled some unique relationships between song and platform attributes that contribute to high streaming success. Some of the correlations to highlight are a strong positive correlation between music in Spotify playlists and music in Apple playlists with a 71% correlation suggesting that songs in Spotify playlists are likely also included in Apple playlists, indicating a shared popularity or playlisting strategy across platforms. A somewhat strong negative correlation between energy and acoustics at -58% indicates how higher-energy songs tend to have lower acoustics.

References

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